31 August 2026 20:37

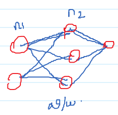
Advanced Data Science

Machine learning	Deep learning
→ Subset of A.I	→ Subset of ML
→ Structured data (table)	→ Structured & unstructured (image, audio, video) Powerful
→ Small dataset	→ Large No of dataset
→ Computing time is less	→ Computing time is more
→ CPU → Sufficient to train the model in fast execution	→ cpu is not sufficient.
→ Kmeans, SVM, DT, ...etc	→ Neural n/ws → ANN (2-3), CNN (≥3+)

How human brain works?

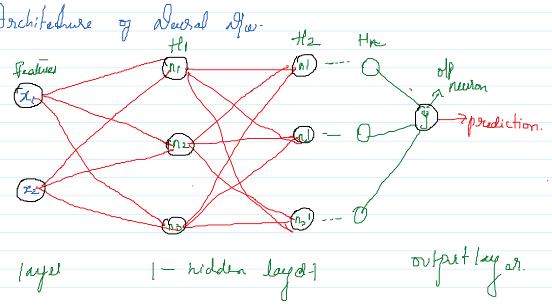
Child → Book

more info & train



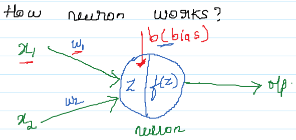
Aritificial brain →
↓
neurons.
↓
Neural n/w.

III Architecture of Neural n/w.



x_1	x_2	Reg y
		1.22
		5.33

IV How neuron works?



$$\boxed{z = x_1w_1 + x_2w_2 + b}$$

$f(z)$ → function of Z

① $f(z)$ → Relu function

$$\boxed{f(z) = \max(z, 0)}$$

Assume

Initial the model will select Random values of w_1 & b

$x_1 = 4 \quad x_2 = 5$

$w_1 = 2 \quad w_2 = 3$

$b = 1$

$z = x_1w_1 + x_2w_2 + b$

$z = 24$

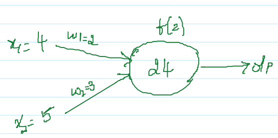
$f(z) = \max(z, 0)$

$\max(24, 0) = 24.$

Eg:- max(25,0)
↓
25

② max(-3,0)
↓
0

Relu → filter the Value



Output layer

Regression.

f(z) → linear Activation function

only 1 neuron

Bi - classifi

	o/p Pred	
f(z) → class 0 → No	1	0
f(z) → class 1 → Yes	0	1

$f(z)$ → Sigmoid $\frac{1}{1+e^{-z}}$ Activation function

Multi-classi (iris - 3 labels)

f(z) → Setosa 0

f(z) → Versi 0

f(z) → Virginica 1

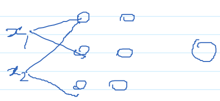
$f(z)$ → Softmax

$f(z) \Rightarrow$ Softmax
Activation funct.

hidden layer $\Rightarrow f(z) \Rightarrow$ Relu

$x_1 \quad x_2 \quad y$

i/p



4 rows $\Rightarrow$ $\boxed{\text{2 rows}}$ ($r_1\, r_2$) B_1 $\boxed{\text{2 rows}}$ ($r_3\, r_4$) B_2

Regularization : Technique used to Reduce Overfitting

Conduct :- $Q_1, Q_2, Q_3 \ldots Q_{10}$ } same $\Rightarrow$

I $\frac{10}{10} \Rightarrow$ Very good $\longrightarrow$ Question are known to shalini

If I conduct test and Questions are new

I $\frac{2}{10} \Rightarrow$ Very bad / not good $\Rightarrow$ New Question

$\therefore$ Shalini performance is very good for Known Question but failed to perform for New/Un known Question.

epoch $\Rightarrow$ one time trained the entire
2 epoch
⋮
n epoch

rand w &b

20 records

$r_1\, r_2\, r_3\, r_4\, r_5$	$r_6 \ldots r_{10}$	$r_{11} \ldots r_{15}$	$r_{16} \ldots r_{20}$
B_1	B_2	B_3	B_4

1 epoch
how many times U have trained entire data

random rows
$r_2, r_3, r_{12}, r_{15}, r_{20}$ B_1 $\quad$ $r_1, r_4, r, 5, r_{16}$ B_2 $\quad$ B_3 $\quad$ B_4

2nd time
2 epoch

Model Works very well for the Trained data.

Model Works very well for the Trained data.

Xtrain → [model] → Ypred. ⇒ good Model

↑ Xtrain (during testing)

MAE

but failed to work for test data.

Xtrain → [ ] → pred ⇒ not good

↑ Xtest (during) / New data (unknown)

Regularization: To reduce Overfitting

Shalini I $\frac{10}{10}$ = good.

II $\frac{3}{10}$ ⇒ $\frac{4}{10}$ $\frac{3}{10}$ $\frac{4}{10}$

by adding the penalty factor.

I L_1 (Lasso)

$$\boxed{\text{Total loss} = \text{Error } (A-P) \text{ MSE} + \text{Penalty factor} + \lambda \sum_{i=1}^{n} W_i}$$

∴ If any Features have less correlation w.r.t y ⇒ remove the Feature

II L_2 (Ridge)

$$\boxed{\text{Total loss} = \text{Error} + \lambda \sum_{i=1}^{n} W^2}$$

∴ F_1 F_2 F_3 F_4 Target
(20%) (80%) (60%) 60%.

F_2 F_4

Randomly Model Selected

W_1, W_2, b.

↓

[batch 1] data

↓

[N/w]

↓

Pred

↓

Error

↓

Error + Penalty ⇒ gradient

↓

$W_{new} = W_{old} - \eta$ gradient

find the updated value of W & b

W_{1new}, W_{2new} & b_{new}

batch 2

w_1 w_2 b ⇒ updated weights

↓ train the model

↓ weight → batch 1

↓ train the ..

↓

updated weight → batches

Sentimental Analysis

I love ML → happy (+ve)

I hate you
-ve

Subjectivity

There are 7 Contients in word — opinion
— Fact

I love this batch

Text → model

Text preprocessing

Numerical (scaling)

cate — Numeri

Gender

F → 0

M → 1

Bag of words

You won 10lakh coupon

Tokenization   Spam

NLP: Natural Lang Processing

① Text preprocessing ⇒ Text → Vector/numbers.